

Data Science

CSE558

Recap

- Fast Matrix Multiplication

$$X = \left\{ \frac{1}{p_k} A(:, k) B(k, :), \quad \text{with probability } p_k \right.$$

$$E \left[\left(AB - \frac{1}{s} \sum_{i=1}^s X_i \right)^2 \right] \leq \frac{\|A\|_F^2 \|B\|_F^2}{s}$$

Overview

- Fast Matrix Multiplication
- Matrix Completion

Sampling

Sampling not helpful

Let $A = B = I_n$.

Then, $\|AB\|_F^2 = n$ and $\|A\|_F^2 \|B\|_F^2 = n^2$

$$E \left[\left(AB - \frac{1}{s} \sum_{i=1}^s X_i \right)^2 \right] \leq \frac{\|A\|_F^2 \|B\|_F^2}{s}$$

To get a useful bound, we need $s > n$.

Sampling

Sampling is helpful

Let A has n identical rows and $A^\top = B$.

Then, $\|AB\|_F^2 = n^2$ and $\|A\|_F^2 \|B\|_F^2 = n^2$

$$E \left[\left(AB - \frac{1}{s} \sum_{i=1}^s X_i \right)^2 \right] \leq \frac{\|A\|_F^2 \|B\|_F^2}{s}$$

By setting $s = \varepsilon^{-1}$, we get a useful bound.

Sampling based Low Rank Approximation

Given a matrix A , find C & W such that, $AI \approx CW$

Is it possible by sampling? No!

We bound $\|A - CW\|_F^2$ by $\|A\|_F^2 \|I\|_F^2 / s = \|A\|_F^2 \frac{n}{s}$

Sampling based Low Rank Approximation

Let $A \in \mathbb{R}^{n \times d}$ (s.t. $\text{rank}(A) = r < d$) and B be r independent rows of A .

$B^T(BB^T)^{-1}B = P$, where P is a projection matrix from the row space of A .

$$AP = A$$

Now, sample s columns of A , form C .

Next, select corresponding s rows of P and form W .

$$E(\|AP - CW\|_2^2) \leq E(\|AP - CW\|_F^2) \leq \frac{\|A\|_F^2 \|P\|_F^2}{\varepsilon} \leq \frac{r}{s} \|A\|_F^2$$

Sampling based Low Rank Approximation

SVD Drawback: Structural Loss.

Use sampling

$$\begin{bmatrix} A \\ n \times m \end{bmatrix} \approx \begin{bmatrix} \text{Sample columns} \\ n \times s \end{bmatrix} \begin{bmatrix} \text{Multiplier} \\ s \times r \end{bmatrix} \begin{bmatrix} \text{Sample rows} \\ r \times m \end{bmatrix}$$

Algorithm

$$\left[\begin{array}{c} \\ \\ A \\ \\ \\ n \times m \end{array} \right] \approx \left[\begin{array}{c} \\ \\ \text{Sample} \\ \text{columns} \\ \\ n \times s \end{array} \right] \left[\begin{array}{c} \text{Multi} \\ \text{plier} \\ s \times r \end{array} \right] \left[\begin{array}{c} \text{Sample rows} \\ r \times m \end{array} \right]$$

Sample k/ε^4 columns and rows from A with respect to their squared L2 norm.

$$\|A - CUR\|_F \leq \|A - A_k\|_F + \varepsilon \|A\|_F$$

Overview

- Fast Matrix Multiplication
- Matrix Completion

Matrix Completion

1	?	2	5	?
2	?	3	?	?
?	?	1	4	?
?	2	?	?	3
?	5	?	4	1
1	?	?	?	?
?	2	?	2	?
4	?	3	?	?
?	4	?	5	?
?	4	?	?	1

X

Solutions

1. Imputation (mean, median, etc)
2. Recommendation: Imputation is the problem

Imputation does not capture correlations among variables.

Matrix Completion

Movies

1	?	2	5	?
2	?	3	?	?
?	?	1	4	?
?	2	?	?	3
?	5	?	4	1
1	?	?	?	?
?	2	?	2	?
4	?	3	?	?
?	4	?	5	?
?	4	?	?	1

Users

Netflix challenge: 17770 users and 480189 movies

Assume missing values are random.

$$Y = \arg \min_{\mathbf{B} \text{ s.t. } \text{rank}(\mathbf{B}) \leq k} \sum_{\text{observed } (j,k)} [\mathbf{X}_{j,k} - \mathbf{B}_{j,k}]^2$$

Matrix Completion

Initialize: Create a complete matrix $\mathbf{M}$ from $\mathbf{X} \in \mathbb{R}^{n \times d}$ such missing values are imputed using mean or median or etc.

Repeat: (1-2) until $\sum_{(i,j) \text{ in observed set}} (\mathbf{M}_{i,j} - \mathbf{X}_{i,j})^2 \leq \tau$

1. $\min_{\mathbf{P} \in \mathbb{R}^{n \times k}, \mathbf{Q} \in \mathbb{R}^{k \times d}} \|\mathbf{M} - \mathbf{PQ}\|_F^2$

2. For every missing entry (i,j) , $\mathbf{M}_{i,j} \leftarrow \mathbf{X}_{i,j}$